The sharp A_p bound for the Hardy–Littlewood maximal operator

An exposition of Lerner's proof of Buckley's theorem

Simon Gustafsson

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1 Introduction

Definition 1 (Weights)**.** *A* weight *is a function $w \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ with $w \geq 0$. For a measurable set $E \subseteq \mathbb{R}^d$ we write $w(E) := \int_E w(x)\,dx$.*

Definition 2 (Weighted L^p spaces)**.** *Let w be a weight and let $1 \leq p < \infty$. We define*

$$L^p_w(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} |f(x)|^p\, w(x)\,dx < \infty \right\},$$

with norm $\|f\|_{L^p_w} := \left(\int_{\mathbb{R}^d} |f(x)|^p\, w(x)\,dx\right)^{1/p}$. Moreover, $L^\infty_w(\mathbb{R}^d)$ is the space of measurable functions f with $\|f\|_{L^\infty_w} := \inf\{t \geq 0 : w(\{|f| > t\}) = 0\} < \infty$.

The central object of this report is the Hardy–Littlewood maximal operator M. Our proofs also use its analogue with respect to the measure $w(x)\,dx$.

Definition 3 (Hardy–Littlewood maximal operator)**.** *Let w be a weight with $w(Q) > 0$ for every cube Q. For a measurable function f and $x \in \mathbb{R}^d$,*

$$M_w f(x) := \sup_{Q \ni x} \frac{1}{w(Q)} \int_Q |f(y)|\, w(y)\,dy,$$

where the supremum is taken over all closed cubes $Q \subset \mathbb{R}^d$ with sides parallel to the axes containing x. All cubes in this report are of this kind. If the supremum is taken only over the cubes $Q(x,r)$, $r > 0$, where $Q(x,r)$ denotes the cube centred at x with side length r, we write M^c_w. If $w \equiv 1$ we drop the subscript and write M and M^c.

It is well known that M is of weak type $(1,1)$ and strong type (p,p) for $1 < p \leq \infty$ [3, Theorem 1.1]. For $1 < p < \infty$ and a weight w we say that M is bounded on $L^p_w(\mathbb{R}^d)$ if there is a constant C such that

$$\|Mf\|_{L^p_w} = \left(\int_{\mathbb{R}^d} (Mf)^p\, w\,dx\right)^{1/p} \leq C \left(\int_{\mathbb{R}^d} |f|^p\, w\,dx\right)^{1/p} = C\,\|f\|_{L^p_w} \tag{1}$$

for all $f \in L^p_w(\mathbb{R}^d)$.

Remark 1 (The zero weight)**.** *If $w = 0$ almost everywhere, then $\|f\|_{L^p_w} = 0$ for every f, so $L^p_w(\mathbb{R}^d)$ is trivial and M is trivially bounded.*

Remark 2 (The endpoint $p = \infty$)**.** *For $p = \infty$, the operator M is bounded on $L^\infty_w(\mathbb{R}^d)$ if and only if $w > 0$ almost everywhere or $w = 0$ almost everywhere. Muckenhoupt [5, Theorem 3] proves this on an interval, and the same argument works in $\mathbb{R}^d$. This is why we only consider $1 < p < \infty$.*

If (1) holds for some weight w, then w cannot be much smaller on one part of a cube than on the rest of it. The following proposition makes this precise, and we prove it in Appendix A.

Proposition 1. *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. If (1) holds with constant C, then for every cube Q we have $\int_Q w^{-\frac{1}{p-1}}\,dx < \infty$ and*

$$\left(\frac{1}{|Q|}\int_Q w\,dx\right)\left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\,dx\right)^{p-1} \le C^p. \tag{2}$$

In particular, $w > 0$ almost everywhere.

This motivates the following definition [5, (7.10)].

Definition 4 (Muckenhoupt A_p weights)**.** *Let $1 < p < \infty$ and let w be a weight. The A_p* characteristic *of w is*

$$[w]_{A_p} := \sup_Q \left(\frac{1}{|Q|}\int_Q w(x)\,dx\right)\left(\frac{1}{|Q|}\int_Q w(x)^{-\frac{1}{p-1}}\,dx\right)^{p-1}, \tag{3}$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$. We say that $w \in A_p$ if $[w]_{A_p} < \infty$.

Remark 3 (The dual weight)**.** *For a weight w and $1 < p < \infty$ we write $\sigma := w^{-\frac{1}{p-1}}$, with $\sigma = \infty$ where $w = 0$, and $\sigma(E) := \int_E \sigma\,dx$ for measurable sets E. With this notation (3) takes the more compact form*

$$[w]_{A_p} = \sup_Q \frac{w(Q)}{|Q|}\left(\frac{\sigma(Q)}{|Q|}\right)^{p-1}.$$

Taking the supremum over all cubes in Proposition 1 gives the following.

Corollary 1. *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. If (1) holds with constant C, then $w \in A_p$ and*

$$[w]_{A_p} \le C^p. \tag{4}$$

Muckenhoupt proved that the condition $w \in A_p$ is also sufficient.

Theorem A (Muckenhoupt [5, Theorem 9])**.** *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. Then M is bounded on $L^p_w(\mathbb{R}^d)$ if and only if $w \in A_p$.*

Theorem A shows that the operator norm of M on $L^p_w(\mathbb{R}^d)$, that is, the smallest constant C in (1), is finite precisely when $w \in A_p$. By Corollary 1, this norm is bounded from below by $[w]_{A_p}^{1/p}$. The natural question is therefore whether it is also bounded from above by a power of $[w]_{A_p}$, and which power is the smallest possible. This was answered by Buckley [1, Theorem 2.5].

Theorem 1 (Buckley). *Let $1 < p < \infty$. There is a constant $c_{p,d}$, depending only on p and d, such that for every $w \in A_p$ and every $f \in L^p_w(\mathbb{R}^d)$,*

$$\|Mf\|_{L^p_w} \le c_{p,d}\,[w]_{A_p}^{\frac{1}{p-1}}\,\|f\|_{L^p_w}.$$

Proposition 2 (Buckley). *Let $1 < p < \infty$ and $\alpha < \frac{1}{p-1}$. Then there is no constant c such that $\|Mf\|_{L^p_w} \le c\,[w]_{A_p}^{\alpha}\,\|f\|_{L^p_w}$ for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$.*

Buckley's proof of Theorem 1 [1, proof of Theorem 2.5], like Muckenhoupt's proof of Theorem A [5, Lemma 5 and Theorem 2], relies on the fact that every A_p weight belongs to $A_{p-\varepsilon}$ for some $\varepsilon > 0$, which is a consequence of the reverse Hölder inequality for A_p weights [3, Theorem 4.5], together with interpolation. Lerner [4, Theorem C] found a short and elementary proof of Theorem 1 which avoids this property, and in this report we present his argument in detail. We complete it with a proof of Proposition 2, based on Buckley's example in $\mathbb{R}$ [1, proof of Theorem 2.5], which we carry out in $\mathbb{R}^d$.

Remark 4 (A proof of Theorem A). *By Corollary 1, which follows from Proposition 1, the condition $w \in A_p$ is necessary for M to be bounded on $L^p_w(\mathbb{R}^d)$, and by Theorem 1 it is sufficient. Hence the proofs in this report also give a proof of Theorem A.*

2 Proofs

2.1 Preliminaries

Lemma 1. *Let Q be a cube with side length $\ell(Q)$ and let $y \in Q$. Then $Q \subseteq Q(y, 2\ell(Q)) \subseteq 3Q$, where $3Q$ denotes the cube with the same centre as Q and side length $3\ell(Q)$.*

Proof. Every $z \in Q$ satisfies $|z_i - y_i| \le \ell(Q)$ for $i = 1, \dots, d$, since both z and y lie in Q. Hence $Q \subseteq Q(y, 2\ell(Q))$. Let c be the centre of Q. Then $|y_i - c_i| \le \frac{1}{2}\ell(Q)$ for $i = 1, \dots, d$, so every $z \in Q(y, 2\ell(Q))$ satisfies

$$|z_i - c_i| \le |z_i - y_i| + |y_i - c_i| \le \ell(Q) + \tfrac{1}{2}\ell(Q) = \tfrac{3}{2}\ell(Q)$$

for $i = 1, \dots, d$, that is, $z \in 3Q$. □

Lemma 2. *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. Then $Mf(x) \le 2^d M^c f(x)$ for every $x \in \mathbb{R}^d$.*

Proof. Let $x \in \mathbb{R}^d$. For a cube Q containing x, Lemma 1 gives $Q \subseteq Q(x, 2\ell(Q))$, and since $|Q(x, 2\ell(Q))| = 2^d|Q|$ we obtain

$$Mf(x) = \sup_{Q \ni x} \frac{1}{|Q|}\int_Q |f(y)|\,dy \le \sup_{Q \ni x} \frac{2^d}{|Q(x, 2\ell(Q))|}\int_{Q(x,2\ell(Q))} |f(y)|\,dy \le 2^d M^c f(x),$$

where the last inequality holds because each $Q(x, 2\ell(Q))$ is one of the cubes in the definition of $M^c f(x)$. □

Lemma 3. *Let $1 < p < \infty$ and let $w \in A_p$ be a weight that is not almost everywhere zero. Then $w > 0$ almost everywhere, and σ is a weight. Moreover $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube Q.*

Proof. Suppose that $w = 0$ on a set E with $|E| > 0$. Since w is not almost everywhere zero, there is also a set F with $|F| > 0$ on which $w > 0$. Choose bounded subsets $E' \subseteq E$ and $F' \subseteq F$ of positive measure, and a cube Q containing both. Then $w(Q) \geq w(F') > 0$, while $\sigma = \infty$ on E', so $\sigma(Q) = \infty$. Hence the quantity in (3) is infinite for this cube, which contradicts $w \in A_p$. Thus $w > 0$ almost everywhere, and in particular $w(Q) > 0$ for every cube Q. Since w is locally integrable, $w < \infty$ almost everywhere, so $\sigma > 0$ almost everywhere and $\sigma(Q) > 0$ for every cube Q. Finally, by the definition of $[w]_{A_p}$,

$$\left(\frac{\sigma(Q)}{|Q|}\right)^{p-1} \leq [w]_{A_p} \frac{|Q|}{w(Q)} < \infty$$

for every cube Q. Since every compact set is contained in a cube, σ is locally integrable. □

Remark 5 (The Vitali covering lemma)**.** *The classical proof of the weak type* $(1,1)$ *estimate for* M *uses the Vitali covering lemma [3, Lemma A.4], which selects disjoint cubes* Q *whose enlargements* $3Q$ *cover the given family. For the measure* $v(x)\,dx$ *this enlargement costs the factor* $v(3Q)/v(Q)$*, which is not bounded uniformly in the weight* v*. The following lemma avoids any enlargement.*

Lemma 4 (Besicovitch covering lemma [2, Theorem 1.1])**.** *There is a constant* N_d*, depending only on* d*, with the following property. Let* $E \subset \mathbb{R}^d$ *be a bounded set and, for each* $x \in E$*, let* $r_x > 0$*. Then there are points* $x_1, x_2, \ldots \in E$ *such that the cubes* $Q(x_j, r_{x_j})$ *cover* E *and satisfy* $\sum_j \chi_{Q(x_j, r_{x_j})} \leq N_d$*.*

Proof. For $y \in \mathbb{R}^d$ we write $|y|_\infty := \max_{1 \leq i \leq d} |y_i|$, so that $y \in Q(x, r)$ if and only if $|y - x|_\infty \leq r/2$. We may assume that E is nonempty, and we let $D := \sup_{x,y \in E} |x - y|_\infty < \infty$. If $r_x \geq 2D$ for some $x \in E$, then $E \subseteq Q(x, r_x)$, and the single point $x_1 := x$ has the required properties. Hence we may assume that $r_x < 2D$ for every $x \in E$.

We choose points $x_1, x_2, \ldots \in E$ inductively. Let $m \geq 0$ and suppose that $x_1, \ldots, x_m$ have been chosen. If $E \setminus \bigcup_{j \leq m} Q(x_j, r_{x_j})$ is empty, we stop. Otherwise we let $s_m := \sup\{r_x : x \in E \setminus \bigcup_{j \leq m} Q(x_j, r_{x_j})\}$ and choose $x_{m+1} \in E \setminus \bigcup_{j \leq m} Q(x_j, r_{x_j})$ with $r_{x_{m+1}} > s_m/2$. If $j < i$, then $x_i \notin Q(x_j, r_{x_j})$, and x_i was also a possible choice when x_j was chosen, so

$$|x_i - x_j|_\infty > \frac{r_{x_j}}{2} \qquad \text{and} \qquad r_{x_i} \leq s_{j-1} < 2r_{x_j}. \tag{5}$$

We show that the cubes $Q(x_j, r_{x_j})$ cover E. This is clear if the selection stops, so suppose that it does not. If $j < i$, then by (5)

$$\frac{r_{x_i}}{8} + \frac{r_{x_j}}{8} < \frac{3r_{x_j}}{8} < |x_i - x_j|_\infty,$$

so the cubes $Q(x_j, r_{x_j}/4)$ are pairwise disjoint. Since $r_{x_j} < 2D$ and $x_j \in E$, they are all contained in $Q(x_1, 4D)$, and therefore $\sum_j (r_{x_j}/4)^d \leq (4D)^d$. In particular $r_{x_j} \to 0$ as $j \to \infty$. If some $x \in E$ were not covered, then x would be a possible choice at every step, so $r_x \leq s_{j-1} < 2r_{x_j}$ for every j, which contradicts $r_{x_j} \to 0$.

It remains to bound the overlap. Fix $z \in \mathbb{R}^d$. The space $\mathbb{R}^d$ is the union of the 2^d closed orthants with vertex z, that is, the sets $\{y : \varepsilon_i(y_i - z_i) \geq 0 \text{ for } i = 1, \ldots, d\}$ with $\varepsilon \in \{-1, 1\}^d$. If y and y' lie in the same orthant, then $y_i - z_i$ and $y'_i - z_i$ have the same sign for every i, and hence

$$|y - y'|_\infty \leq \max\{|y - z|_\infty,\ |y' - z|_\infty\}. \tag{6}$$

Fix one orthant and let J be the set of indices i such that x_i lies in this orthant and $z \in Q(x_i, r_{x_i})$. Suppose that J is nonempty and let j be its smallest element. Let $i \in J$ with $i > j$. Since $|x_j - z|_\infty \le r_{x_j}/2$, (5) and (6) give

$$\frac{r_{x_j}}{2} < |x_i - x_j|_\infty \le \max\{|x_i - z|_\infty,\ |x_j - z|_\infty\} = |x_i - z|_\infty \le \frac{r_{x_i}}{2},$$

so $r_{x_i} > r_{x_j}$, and by (5) also $|x_i - z|_\infty \le r_{x_i}/2 < r_{x_j}$. Moreover, if $i, i' \in J$ with $i > i' > j$, then (5) gives $|x_i - x_{i'}|_\infty > r_{x_{i'}}/2 > r_{x_j}/2$. Hence the cubes $Q(x_i, r_{x_j}/2)$ with $i \in J$ and $i > j$ are pairwise disjoint, and they are contained in $Q(z, 5r_{x_j}/2)$. Comparing volumes, there are at most 5^d such indices i, so J has at most $1 + 5^d$ elements. Summing over the 2^d orthants, z lies in at most $2^d(1+5^d)$ of the cubes $Q(x_j, r_{x_j})$, and the lemma holds with $N_d := 2^d(1 + 5^d)$. □

Lemma 5. *Let v be a weight with $v(Q) > 0$ for every cube Q, and let f be a measurable function such that $|f|\,v$ is locally integrable. Then $M_v^c f$ is measurable, and*

$$v\big(\{x \in \mathbb{R}^d : M_v^c f(x) > \lambda\}\big) \le \frac{N_d}{\lambda}\,\|f\|_{L^1_v}$$

for every $\lambda > 0$, where N_d is the constant in Lemma 4.

Proof. Fix $\lambda > 0$, and call a cube $Q(x, r)$ *good* if

$$\int_{Q(x,r)} |f(y)|\,v(y)\,dy > \lambda\, v(Q(x,r)). \tag{7}$$

By the definition of M_v^c, we have $M_v^c f(x) > \lambda$ if and only if $Q(x, r)$ is good for some $r > 0$. For fixed $r > 0$, both sides of (7) are continuous functions of x. Indeed, if $x' \to x$, then $\chi_{Q(x',r)} \to \chi_{Q(x,r)}$ outside the boundary of $Q(x, r)$, which has measure zero, and $\chi_{Q(x',r)} \le \chi_{Q(x,r+2)}$ whenever $|x' - x| \le 1$, so the claim follows by dominated convergence since $|f|\,v$ and v are integrable on $Q(x, r+2)$. Hence the set U_r of all x for which $Q(x, r)$ is good is open, and so is $\{M_v^c f > \lambda\} = \bigcup_{r>0} U_r$. Since $\lambda > 0$ was arbitrary and $M_v^c f \ge 0$, the function $M_v^c f$ is measurable.

For every positive integer k let $E_k := \{x : |x| < k \text{ and } M_v^c f(x) > \lambda\}$. For each $x \in E_k$ we choose $r_x > 0$ such that $Q(x, r_x)$ is good. Since E_k is bounded, Lemma 4 gives points $x_1, x_2, \ldots \in E_k$ such that the cubes $Q(x_j, r_{x_j})$ cover E_k and satisfy $\sum_j \chi_{Q(x_j, r_{x_j})} \le N_d$. Using (7) for each $Q(x_j, r_{x_j})$ and monotone convergence to interchange sum and integral, we obtain

$$\begin{aligned} v(E_k) &\le \sum_j v\big(Q(x_j, r_{x_j})\big) \le \frac{1}{\lambda} \sum_j \int_{Q(x_j, r_{x_j})} |f|\,v\,dy \\ &= \frac{1}{\lambda} \int_{\mathbb{R}^d} \Big(\sum_j \chi_{Q(x_j, r_{x_j})}\Big) |f|\,v\,dy \le \frac{N_d}{\lambda}\,\|f\|_{L^1_v}. \end{aligned}$$

Since $E_1 \subseteq E_2 \subseteq \cdots$ and $\bigcup_k E_k = \{M_v^c f > \lambda\}$, continuity of the measure $v(x)\,dx$ from below gives the claim. □

Lemma 6. *Let v be a weight with $v(Q) > 0$ for every cube Q, and let $1 < q < \infty$. Then there is a constant $C_{q,d}$, depending only on q and d, such that*

$$\|M_v^c f\|_{L^q_v} \le C_{q,d}\,\|f\|_{L^q_v}$$

for all $f \in L^q_v(\mathbb{R}^d)$.

Proof. If $f = f_1 + f_2$ with $f_1 \in L^1_v(\mathbb{R}^d)$ and $f_2 \in L^\infty_v(\mathbb{R}^d)$, then $\int_Q |f|\, v\, dy \le \|f_1\|_{L^1_v} + \|f_2\|_{L^\infty_v}\, v(Q) < \infty$ for every cube Q. Hence M^c_v is defined on $L^1_v(\mathbb{R}^d) + L^\infty_v(\mathbb{R}^d)$, and $M^c_v f$ is measurable by Lemma 5. Since $|f+h| \le |f| + |h|$, the operator M^c_v is sublinear.

By Lemma 5, $v(\{M^c_v f > \lambda\}) \le \frac{N_d}{\lambda}\|f\|_{L^1_v}$ for every $f \in L^1_v(\mathbb{R}^d)$ and $\lambda > 0$. Moreover, if $f \in L^\infty_v(\mathbb{R}^d)$, then $|f| \le \|f\|_{L^\infty_v}$ outside a set of v-measure zero, so every average in the definition of $M^c_v f$ is at most $\|f\|_{L^\infty_v}$, and $\|M^c_v f\|_{L^\infty_v} \le \|f\|_{L^\infty_v}$. The Marcinkiewicz interpolation theorem [3, Theorem A.6], applied on the measure space $(\mathbb{R}^d, v(x)\,dx)$ with exponents 1 and ∞, now gives the claim with a constant depending only on q, N_d and 1, and hence only on q and d. □

2.2 Proof of Theorem 1

Proof of Theorem 1. Let $1 < p < \infty$, let $w \in A_p$ and let $f \in L^p_w(\mathbb{R}^d)$. If $w = 0$ almost everywhere, both sides of the inequality vanish, so we may assume that w is not almost everywhere zero. By Lemma 3, $w > 0$ almost everywhere and σ is a weight with $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube Q. In particular, the operators M^c_w and M^c_σ are defined.

We first prove a pointwise estimate for Mf. Fix $x \in \mathbb{R}^d$ and $r > 0$, let $Q := Q(x,r)$, and let $3Q$ be as in Lemma 1. Multiplying and dividing by $w(Q)\,\sigma(3Q)^{p-1}$, we obtain

$$\begin{aligned}\left(\frac{1}{|Q|}\int_Q |f(y)|\,dy\right)^{p-1} &= \frac{1}{|Q|^{p-1}}\left(\int_Q |f(y)|\,dy\right)^{p-1}\\ &= \frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p}\cdot\frac{|Q|}{w(Q)}\cdot\frac{1}{\sigma(3Q)^{p-1}}\left(\int_Q |f(y)|\,dy\right)^{p-1}\\ &= \frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p}\cdot\frac{|Q|}{w(Q)}\left(\frac{1}{\sigma(3Q)}\int_Q |f(y)|\,dy\right)^{p-1}.\end{aligned} \tag{8}$$

Since $w(Q) \le w(3Q)$ and $|3Q| = 3^d|Q|$, the first factor on the right-hand side satisfies

$$\frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p} \le 3^{dp}\left(\frac{1}{|3Q|}\int_{3Q} w(y)\,dy\right)\left(\frac{1}{|3Q|}\int_{3Q}\sigma(y)\,dy\right)^{p-1} \le 3^{dp}\,[w]_{A_p}.$$

For the second factor, let $y \in Q$. By Lemma 1, $Q \subseteq Q(y, 2\ell(Q)) \subseteq 3Q$, so

$$\begin{aligned}\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz &= \frac{1}{\sigma(3Q)}\int_Q |f(z)|\,\sigma(z)^{-1}\,\sigma(z)\,dz\\ &\le \frac{1}{\sigma(Q(y,2\ell(Q)))}\int_{Q(y,2\ell(Q))} |f(z)|\,\sigma(z)^{-1}\,\sigma(z)\,dz \le M^c_\sigma\big(|f|\sigma^{-1}\big)(y).\end{aligned}$$

This holds for every $y \in Q$, and therefore

$$\begin{aligned}\frac{|Q|}{w(Q)}\left(\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz\right)^{p-1} &= \frac{1}{w(Q)}\int_Q\left(\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz\right)^{p-1} dy\\ &\le \frac{1}{w(Q)}\int_Q \Big(M^c_\sigma\big(|f|\sigma^{-1}\big)(y)\Big)^{p-1} w(y)^{-1}\,w(y)\,dy\\ &\le M^c_w\Big(\big(M^c_\sigma(|f|\sigma^{-1})\big)^{p-1} w^{-1}\Big)(x),\end{aligned}$$

where the last inequality holds because $Q = Q(x,r)$. Inserting both bounds into (8), taking the supremum over $r > 0$ and applying Lemma 2, we arrive at

$$Mf(x) \le 2^d\, 3^{\frac{dp}{p-1}}\,[w]_{A_p}^{\frac{1}{p-1}}\Big(M^c_w\Big(\big(M^c_\sigma(|f|\sigma^{-1})\big)^{p-1} w^{-1}\Big)(x)\Big)^{\frac{1}{p-1}}. \tag{9}$$

We now take norms. Since $1 - \frac{p}{p-1} = -\frac{1}{p-1}$, we have $w^{1-\frac{p}{p-1}} = \sigma$, and similarly $\sigma^{1-p} = w$. The second identity gives

$$\big\| |f|\sigma^{-1} \big\|_{L^p_\sigma}^p = \int_{\mathbb{R}^d} |f(x)|^p \, \sigma(x)^{-p} \, \sigma(x) \, dx = \int_{\mathbb{R}^d} |f(x)|^p \, w(x) \, dx = \|f\|_{L^p_w}^p,$$

so $|f|\sigma^{-1} \in L^p_\sigma(\mathbb{R}^d)$. Using the first identity and Lemma 6 with $v = \sigma$ and $q = p$, we obtain

$$\begin{aligned} \int_{\mathbb{R}^d} \Big(\big(M^c_\sigma(|f|\sigma^{-1})(x) \big)^{p-1} w(x)^{-1} \Big)^{\frac{p}{p-1}} w(x) \, dx &= \int_{\mathbb{R}^d} \big(M^c_\sigma(|f|\sigma^{-1})(x) \big)^p \, \sigma(x) \, dx \\ &\le C_{p,d}^p \, \|f\|_{L^p_w}^p, \end{aligned}$$

so the function $\big(M^c_\sigma(|f|\sigma^{-1}) \big)^{p-1} w^{-1}$ lies in $L^{p/(p-1)}_w(\mathbb{R}^d)$. Finally, we raise (9) to the power p, integrate against w, and apply Lemma 6 with $v = w$ and $q = \frac{p}{p-1}$. Together with the previous estimate this gives

$$\|Mf\|_{L^p_w}^p \le \Big(2^d \, 3^{\frac{dp}{p-1}} \Big)^p \, [w]_{A_p}^{\frac{p}{p-1}} \, C_{\frac{p}{p-1},d}^{\frac{p}{p-1}} \, C_{p,d}^p \, \|f\|_{L^p_w}^p.$$

Taking p-th roots, we conclude that

$$\|Mf\|_{L^p_w} \le c_{p,d} \, [w]_{A_p}^{\frac{1}{p-1}} \, \|f\|_{L^p_w}, \qquad c_{p,d} := 2^d \, 3^{\frac{dp}{p-1}} \, C_{\frac{p}{p-1},d}^{\frac{1}{p-1}} \, C_{p,d},$$

and $c_{p,d}$ depends only on p and d. □

2.3 Sharpness of the exponent

Proof of Proposition 2. Let $1 < p < \infty$ and $\alpha < \frac{1}{p-1}$, and suppose, for contradiction, that there is a constant c such that

$$\|Mf\|_{L^p_w} \le c \, [w]_{A_p}^\alpha \, \|f\|_{L^p_w} \tag{10}$$

for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$. For every weight w that is not almost everywhere zero and every cube Q, Hölder's inequality with exponents p and $\frac{p}{p-1}$ gives

$$1 = \frac{1}{|Q|} \int_Q w^{\frac{1}{p}} \, w^{-\frac{1}{p}} \, dy \le \left(\frac{1}{|Q|} \int_Q w \, dy \right)^{\frac{1}{p}} \left(\frac{1}{|Q|} \int_Q w^{-\frac{1}{p-1}} \, dy \right)^{\frac{p-1}{p}},$$

so $[w]_{A_p} \ge 1$. Hence (10) remains true if α is replaced by $\max\{\alpha, 0\}$, and we may assume that $0 \le \alpha < \frac{1}{p-1}$.

For $0 < \delta < 1$ we consider the weight $w_\delta(x) := |x|^{(d-\delta)(p-1)}$, for which $w_\delta^{-\frac{1}{p-1}}(x) = |x|^{\delta - d}$. By polar coordinates,

$$\int_{\{|y|<r\}} |y|^{\delta-d} \, dy = \omega_d \, \frac{r^\delta}{\delta} \qquad \text{for every } r > 0, \tag{11}$$

where ω_d denotes the surface measure of the unit sphere in $\mathbb{R}^d$.

We first show that $w_\delta \in A_p$ with

$$[w_\delta]_{A_p} \le C \, \delta^{1-p}, \tag{12}$$

where C depends only on p and d. The function w_δ is continuous and positive outside the origin, so it is a weight that is not almost everywhere zero. Write $a := (d-\delta)(p-1) \geq 0$, and let $Q := Q(c,\ell)$ be a cube. Every $y \in Q$ satisfies $|y-c| \leq \frac{\sqrt{d}}{2}\,\ell$.

Suppose first that $|c| \leq \sqrt{d}\,\ell$. Then every $y \in Q$ satisfies $|y| \leq R\ell$ with $R := \frac{3\sqrt{d}}{2}$. Since $a \geq 0$, this gives $w_\delta \leq (R\ell)^a$ on Q, and by (11),

$$\frac{1}{|Q|}\int_Q |y|^{\delta-d}\,dy \leq \frac{1}{\ell^d}\int_{\{|y|<R\ell\}} |y|^{\delta-d}\,dy = \frac{\omega_d\,R^\delta\,\ell^{\delta-d}}{\delta}.$$

Since $a + (\delta-d)(p-1) = 0$, we obtain

$$\left(\frac{1}{|Q|}\int_Q w_\delta\,dy\right)\left(\frac{1}{|Q|}\int_Q |y|^{\delta-d}\,dy\right)^{p-1} \leq (R\ell)^a\left(\frac{\omega_d\,R^\delta\,\ell^{\delta-d}}{\delta}\right)^{p-1} = R^{d(p-1)}\,\omega_d^{p-1}\,\delta^{1-p}.$$

Suppose instead that $|c| > \sqrt{d}\,\ell$. Then $|y-c| < \frac{1}{2}|c|$, so $\frac{1}{2}|c| \leq |y| \leq \frac{3}{2}|c|$ for every $y \in Q$. Hence $w_\delta \leq \left(\frac{3}{2}|c|\right)^a$ and $|y|^{\delta-d} \leq \left(\frac{1}{2}|c|\right)^{\delta-d}$ on Q, and

$$\left(\frac{1}{|Q|}\int_Q w_\delta\,dy\right)\left(\frac{1}{|Q|}\int_Q |y|^{\delta-d}\,dy\right)^{p-1} \leq \left(\tfrac{3}{2}\right)^a 2^{(d-\delta)(p-1)} = 3^a \leq 3^{d(p-1)} \leq 3^{d(p-1)}\,\delta^{1-p}.$$

Taking the supremum over all cubes proves (12) with $C := \max\{R^{d(p-1)}\omega_d^{p-1},\, 3^{d(p-1)}\}$.

Next we construct a function for which M is large in $L^p_{w_\delta}$. Let $f_\delta(x) := |x|^{\delta-d}\chi_{Q(0,2)}(x)$. Then $f_\delta \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, and since $Q(0,2) \subseteq \{|y| < 2\sqrt{d}\}$, (11) gives

$$\|f_\delta\|^p_{L^p_{w_\delta}} = \int_{Q(0,2)} |y|^{(\delta-d)p}\,|y|^{(d-\delta)(p-1)}\,dy = \int_{Q(0,2)} |y|^{\delta-d}\,dy \leq \omega_d\,\frac{(2\sqrt{d})^\delta}{\delta} < \infty.$$

Moreover $\|f_\delta\|_{L^p_{w_\delta}} > 0$, so $f_\delta \in L^p_{w_\delta}(\mathbb{R}^d)$ is not the zero element. Let $x \in Q(0,2)$ with $x \neq 0$, and put $r := \max_{1\leq i\leq d}|x_i|$, so that $0 < r \leq |x|$. The cube $Q(0,2r)$ contains x, is contained in $Q(0,2)$, and contains the ball $\{|y| < r\}$. By the definition of M and (11),

$$Mf_\delta(x) \geq \frac{1}{(2r)^d}\int_{Q(0,2r)} |y|^{\delta-d}\,dy \geq \frac{1}{(2r)^d}\int_{\{|y|<r\}} |y|^{\delta-d}\,dy = \frac{\omega_d}{2^d\delta}\,r^{\delta-d} \geq \frac{\omega_d}{2^d\delta}\,|x|^{\delta-d},$$

where the last inequality holds because $\delta - d < 0$ and $r \leq |x|$. Thus $Mf_\delta \geq \frac{\omega_d}{2^d\delta} f_\delta$ almost everywhere on $Q(0,2)$, and this also holds outside $Q(0,2)$, where $f_\delta = 0$. Consequently

$$\|Mf_\delta\|_{L^p_{w_\delta}} \geq \frac{\omega_d}{2^d\delta}\,\|f_\delta\|_{L^p_{w_\delta}}. \tag{13}$$

Finally we combine these estimates. Applying (10) to w_δ and f_δ, and using (13), (12) and $\alpha \geq 0$, we obtain

$$\frac{\omega_d}{2^d\delta}\,\|f_\delta\|_{L^p_{w_\delta}} \leq \|Mf_\delta\|_{L^p_{w_\delta}} \leq c\,[w_\delta]^\alpha_{A_p}\,\|f_\delta\|_{L^p_{w_\delta}} \leq c\,C^\alpha\,\delta^{\alpha(1-p)}\,\|f_\delta\|_{L^p_{w_\delta}}.$$

Dividing by $\|f_\delta\|_{L^p_{w_\delta}} \in (0,\infty)$ and multiplying by δ gives

$$\frac{\omega_d}{2^d} \leq c\,C^\alpha\,\delta^{1-\alpha(p-1)} \qquad \text{for every } 0 < \delta < 1.$$

Since $\alpha < \frac{1}{p-1}$, the exponent $1-\alpha(p-1)$ is positive, so the right-hand side tends to 0 as $\delta \to 0$, while the left-hand side is a positive constant independent of δ. This is a contradiction, and the proof is complete. □

A Proof of Proposition 1

We follow the proof of [3, Theorem 1.7].

Proof. We first show that $w > 0$ almost everywhere. Let Q be a cube and let $S \subseteq Q$ be a measurable set. For $x \in Q$ we have $M\chi_S(x) \geq |S|/|Q|$, so (1) applied to $f = \chi_S$ gives

$$\left(\frac{|S|}{|Q|}\right)^p w(Q) \leq C^p\, w(S).$$

Since w is not almost everywhere zero, every bounded set S is contained in a cube Q with $w(Q) > 0$, so $w(S) = 0$ implies $|S| = 0$. Applying this to the sets $\{w = 0\} \cap \{|x| < k\}$, $k = 1, 2, \ldots$, shows that $w > 0$ almost everywhere. In particular $w(Q) > 0$ for every cube Q.

Now let Q be a cube, let $\varepsilon > 0$ and let $f := (w+\varepsilon)^{-\frac{1}{p-1}}\,\chi_Q$. Since $w \leq w + \varepsilon$,

$$(w+\varepsilon)^{-\frac{p}{p-1}}\, w \leq (w+\varepsilon)^{-\frac{p}{p-1}}\,(w+\varepsilon) = (w+\varepsilon)^{-\frac{1}{p-1}} \leq \varepsilon^{-\frac{1}{p-1}},$$

so $f \in L^p_w(\mathbb{R}^d)$. Moreover, Q is a cube containing every $x \in Q$, so $Mf(x) \geq \frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy$ for $x \in Q$, and therefore

$$\begin{aligned}
\left(\frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy\right)^p w(Q) &= \int_Q \left(\frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy\right)^p w(x)\,dx \\
&\leq \int_Q (Mf)^p\, w\,dx \\
&\leq \|Mf\|^p_{L^p_w} \\
&\leq C^p\,\|f\|^p_{L^p_w} \\
&= C^p \int_Q (w+\varepsilon)^{-\frac{p}{p-1}}\, w\,dx \\
&\leq C^p \int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dx.
\end{aligned}$$

Since w is locally integrable, $w < \infty$ almost everywhere, so $(w+\varepsilon)^{-\frac{1}{p-1}} > 0$ almost everywhere and the integral on the right-hand side is positive. Dividing by it we obtain

$$\frac{w(Q)}{|Q|}\left(\frac{1}{|Q|}\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy\right)^{p-1} \leq C^p.$$

Let σ be as in Remark 3. As $\varepsilon \to 0$, $(w+\varepsilon)^{-\frac{1}{p-1}}$ increases to σ, so by monotone convergence $\int_Q (w+\varepsilon)^{-\frac{1}{p-1}}\,dy \to \sigma(Q)$, and (2) follows. Since $w(Q) > 0$, this also shows that $\sigma(Q) < \infty$. □

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